

Case Ace v2.0 Cost Estimate & Economic Model

Document Reference: CAW-FIN-EST-2026-01
System: Case Ace v2.0 Client Consultation Recording & Note Drafting System
Organisation: Citizens Advice Wandsworth (CAW)
Target Audience: Board of Trustees, Finance & Audit Committee, Senior Leadership Team
Effective Date: 2026-09-02
Status: Formally Approved Financial Model
Classification: Official / Financial Planning

1. Executive Financial Summary

This document presents the detailed cost estimate and financial model for Case Ace v2.0 across two operational phases:

The 6-Week Pilot Phase (15 advisers, 250 validated client consultations).
The Full Bureau Roll-Out Phase (75 advisers/volunteers, 2,750 consultations/month).

Due to Case Ace’s client-side in-memory architecture (utilising local in-browser Whisper WASM for zero-cost speech transcription and compact Gemini 1.5 Flash structured drafting in London europe-west2), the operating costs are exceptionally low, delivering unprecedented value for money.

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HEADLINE FINANCIAL METRICS SUMMARY			
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Metric		Pilot Phase (6 Weeks)	Full Roll-Out (Month)
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Active Adviser / Caseworker Cohort		15 Advisers	75 Advisers & Vols
Total Casenotes Generated		250 Consultations	2,750 Notes / Month
Total Operational Running Cost		**£81.66 (Total)**	**£217.65 / Month**
Monthly Operating Cost		**£54.44 / Month**	**£217.65 / Month**
Cost Per Casenote		**£0.33 / Note**	**£0.079 (7.9p) /
Note**			
Cost Per Adviser		**£3.63 / Adv / Month**	**£2.90 / Adv /
Month**			
Caseworker Time Saved		62.5 Hours	687.5 Hours / Month
Capacity Value Unlocked (@ £18.50/hr)		£1,156.25	£12,718.75 / Month
Return on Investment (ROI Ratio)		**14.2 : 1**	**58.4 : 1**
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2. Granular Unit Cost Breakdown

2.1. Google Cloud Vertex AI (Gemini 1.5 Flash - europe-west2 London)

Input Token Pricing: \$0.075 per 1,000,000 tokens (\$\approx \text{\text{£}0.060} / \text{\text{1M tokens}}\$).
Output Token Pricing: \$0.300 per 1,000,000 tokens (\$\approx \text{\text{£}0.240} / \text{\text{1M tokens}}\$).
Tokens Per Consultation Session:
System Prompt + Context + Tokenised Transcript (\$\approx 4,500\text{\text{ words}}\$): \$\approx 6,000\text{\text{ input tokens}}\$.
Cost per input prompt: \$6,000 \times \frac{\text{\text{£}0.060}}{1,000,000} = \text{\text{£}0.00036}\$.
Generated AQS Case Note (\$\approx 800\text{\text{ words}}\$): \$\approx 1,100\text{\text{ output tokens}}\$.
Cost per output draft: \$1,100 \times \frac{\text{\text{£}0.240}}{1,000,000} = \text{\text{£}0.000264}\$.
Total Vertex AI Gemini 1.5 Flash Cost Per Casenote: £0.00062 (\$\approx \text{\text{£}0.062} / \text{\text{pence}}\$).

2.2. Google Cloud Speech-to-Text (Cloud STT v2 / Chirp 2 - London europe-west2)

Primary ASR Engine: Local in-browser Whisper WASM (\$\text{\text{£}0.00}\$ API / compute cost).
Dual-Pass / Difficult Audio Secondary Pass: Cloud STT v2 invoked selectively when local acoustic confidence is \$< 85\%\$ or strong overlapping speech occurs.
Pricing: Standard Model = \$0.016 / \text{\text{minute}} \approx \text{\text{£}0.0128} / \text{\text{minute}}\$.
Average Consultation Duration: 45 minutes of audio \$\rightarrow \text{\text{£}0.58}\$ per full-length cloud transcription.
Blended Rate Assumptions:

Steady-State Roll-Out (10% Cloud STT fallback): $\$0.10 \times \text{\text{£}0.58} = \text{\textbf{\text{£}0.058 / casenote}}\$$.

$$\frac{687.5 \text{ hours / month}}{160 \text{ working hours / month}} = \mathbf{4.3 \text{ Full-Time Equivalent Advisers}}$$

Unlocked}}\$

6.2. Direct Economic Value Delivered

At an average adviser employment cost of £18.50 / hour (salary + NI + pension + overheads):

$$\text{Monthly Capacity Value} = 687.5 \text{ hours} \times \text{£18.50} = \text{£12,718.75 / month} \quad (\text{£152,625.00 / year})$$

Net Financial Return on Investment (ROI):

$$\text{ROI Ratio} = \frac{\text{Capacity Value Unlocked}}{\text{Monthly Operating Cost}} = \frac{\text{£12,718.75}}{\text{£217.65}} = 58.4 : 1$$

[!TIP]

Summary Recommendation for the Board of Trustees:

For an investment of less than £220 per month (\$\approx 8 \text{ pence per client}\$), Citizens Advice Wandsworth unlocks over 685 hours of monthly casework capacity—the equivalent of gaining 4.3 additional full-time advisers dedicated to helping vulnerable clients across the borough.